

On the reliability of the inversion of the Arrhenius rate law

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A numerical test of the reliability of the Slater-Forst procedure for the inversion of the Arrhenius rate law is presented using the theoretical data for the reactions $\text{N}_2\text{O} \rightarrow \text{N}_2 + \text{O}$ and $\text{CO}_2 \rightarrow \text{CO} + \text{O}$ reported previously; the test results are positive. The sensitivity of the procedure to variations in the Arrhenius parameters is also examined.

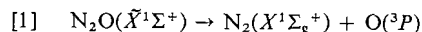
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Faisant appel à des données théoriques relatives aux réactions $\text{N}_2\text{O} \rightarrow \text{N}_2 + \text{O}$ et $\text{CO}_2 \rightarrow \text{CO} + \text{O}$ rapportées antérieurement, on présente une méthode numérique permettant de mettre à l'épreuve l'exactitude de la méthode de Slater-Forst pour l'inversion de la loi de vitesse d'Arrhénius: les résultats de nos essais sont positifs. On examine aussi la sensibilité de la méthode aux variations des paramètres d'Arrhénius.

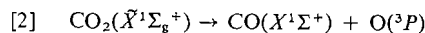
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I. Introduction

In an earlier publication (1), the microscopic reaction probability functions $k(E)$ for the spin-forbidden reactions



and



were calculated in the framework of the theory of radiationless transitions. The calculation was two-dimensional, incorporating the two stretching degrees of freedom in a modified form of Rosen's relative co-ordinate system (2); also, all of the microscopic parameters required for the calculation were derived from spectroscopic data, and there were no adjustable parameters in the formulation. The computed $k(E)$ functions were of "kinetic accuracy": for both reactions, the high-pressure reaction rates calculated from the $k(E)$ functions were in reasonable agreement with experiment (3-5). The pre-exponential factor and activation energy were also computed: they vary only moderately with temperature, except at very low temperature for N_2O . Indeed, over a long range of temperature, the reaction rates may be very well approximated by a strict Arrhenius expression, viz.

$$[3] \quad k_\infty(T) \doteq A_\infty \exp(-E_\infty/RT)$$

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with $A_\infty = 1.41 \times 10^{12} \text{ s}^{-1}$ and $E_\infty = 65.10 \text{ kcal mol}^{-1}$ for reaction [1] and $A_\infty = 1.80 \times 10^{12} \text{ s}^{-1}$ and $E_\infty = 127.3 \text{ kcal mol}^{-1}$ for reaction [2] (1).

The inversion procedure of (experimental) high-pressure rate laws such as eq. [3] to obtain the microscopic reaction probability function $k(E)$ was first discussed by Slater (6) and was later re-derived and developed for unimolecular fall-off calculations by Forst (7). Despite the success to date (8) of the Slater-Forst procedure in unimolecular rate calculations, concerns have been voiced in the past regarding its validity: (i) that the experimental rate law is necessarily approximate; (ii) that the inversion procedure may be very sensitive to the fine structure of $k_\infty(T)$, and the loss of the latter through experimental scatter precludes obtaining even the overall shape of the $k(E)$ function; (iii) that (in the case of the strict Arrhenius rate law, eq. [3]), E_∞ is not equal to E_0 , the critical energy of reaction; and that (iv) the $k(E)$ function is expected to be very sensitive to the assumed Arrhenius parameters, particularly near the threshold. Regarding (i), the $k(E)$ function obtained by the inversion of an approximate rate law is necessarily approximate; the question is how approximate is the function? As to (ii), we have argued previously (8) using the linearity property of Laplace Transform, that the loss of the fine-structure in $k_\infty(T)$ would mean a corresponding loss in the fine-structure, but not the overall shape, of $k(E)$ in the inversion procedure. Case (iii) actually arises out of an over-interpretation of eq. [3]: insofar as the inversion procedure is concerned, both A_∞ and E_∞ are empirical quantities and no assumption is made that E_∞ is equal to E_0 . At this juncture, we note that

no realistic kinetic data (either theoretical or experimental) of $k(E)$ and $k_\infty(T)$ yet exist which may be used to examine (iv) in particular and to investigate the validity of the Slater-Forst procedure in general: hence, the present work using the $k(E)$ and $k_\infty(T)$ data in ref. 1.

II. Theory

We begin with an outline of the relevant theoretical results.

We define, for a discrete reactive state i , g_i as its degeneracy, ε_i as its energy, and k_i as its microscopic decay rate constant to product. The infinite-pressure rate constant $k_\infty(\beta)$ at inverse temperature β ($= 1/kT$) is then (8)

$$[4] \quad k_\infty(\beta) = \sum_j F(E_j) \exp(-\beta E_j)/Q(\beta)$$

where $Q(\beta)$ is the molecular partition function, the j -summation is over all energy levels, and

$$[5] \quad F(E_j) = \sum_i k_i g_i$$

the i -summation being over states at energy E_j . $F(E_j)$ may thus be obtained as the inverse Laplace Transform of $k_\infty(\beta)Q(\beta)$, whence

$$[6] \quad F(E_j) = \mathcal{L}^{-1}\{k_\infty(\beta)Q(\beta)\}$$

and, if the rate law is strict Arrhenius in form, i.e., eq. [3],

$$[7] \quad F(E_j) = A_\infty P(E_j - E_\infty)$$

where $P(E_j)$ is the state-degeneracy at energy E_j .

Equations [4] and [7] may be compared with their corresponding solutions for a continuous system, where

$$[8] \quad k_\infty(\beta) = \int_0^\infty f(E) \exp(-\beta E) dE/Q(\beta)$$

$$[9] \quad f(E) = \rho(E)k(E)$$

and, in the case of the strict Arrhenius rate law,

$$[10] \quad f(E) = A_\infty \rho(E - E_\infty)$$

$\rho(E)$ being the state-density at energy E and $k(E)$ the reaction probability function.

From a computational standpoint, eqs. [4]–[7] and eqs. [8]–[10] are equivalent.² This is because in unimolecular rate calculations, the reactant level spectrum is grouped into energy grids and the relevant integrations are carried out by stepwise

summations over the energy grids. Thus, the quantity of computational interest is

$$[11a] \quad \phi_k = \int_{E_k}^{E_{k+1}} f(E) dE/\Delta E$$

where ΔE is the grid size and E_k is the energy of the k th grid. Equivalently,

$$[11b] \quad \phi_k = \sum F(E_j)/\Delta E$$

where the summation is over energy levels between E_k and E_{k+1} . It is obvious that at high energy ($E_k \gg E_\infty$), eqs. [11a] and [11b] are very similar and ϕ_k varies only slightly with the assumed grid size ΔE . At low energy ($E_k \gtrsim E_\infty$) on the other hand, the behaviour of ϕ_k may be expected to depend more strongly on ΔE . In particular, when ΔE becomes comparable with (or smaller than) the mean energy spacing near the zero point energy, the non-uniform distribution of energy levels there would imply, according to [7], a highly oscillatory distribution of ϕ_k which varies drastically with ΔE . However, this constitutes no serious problem in practical applications, cf. Section IV.

III. Reaction Probability Function ϕ_k

In the theoretical calculation (1) of the microscopic reaction probability, the reactant state is characterised, in the notation of ref. 1, by $(v_{b1}, v_{b2}, v_{\text{bend}}, J, M_J)$ where v_{b1} and v_{b2} refer to the two stretching co-ordinates; v_{bend} denotes the doubly-degenerate bending quantum number, and J and M_J the rotational quantum numbers. The reaction probability $k_{v_{b1} v_{b2} v_{\text{bend}} J M_J}$ from the reactant state is assumed to be independent of v_{bend}, J , and M_J , whence

$$[12] \quad k_{v_{b1} v_{b2} v_{\text{bend}} J M_J} = k_{v_{b1} v_{b2} 0 0 0}$$

with

$$[13] \quad k_{v_{b1} v_{b2} 0 0 0} \equiv k_{v_{b1} v_{b2}} = \sum k_{b, v_{b1}, v_{b2}}^{u, \varepsilon_{u1}, v_{u2}}$$

where $k_{b, v_{b1}, v_{b2}}^{u, \varepsilon_{u1}, v_{u2}}$ is the microscopic reaction probability from the bound (b) reactant state (v_{b1}, v_{b2}) to the unbound (u) product state $(\varepsilon_{u1}, v_{u2})$ and the summation is over all accessible product states $(\varepsilon_{u1}, v_{u2})$. As discussed in ref. 1, the reaction probability $k_{v_{b1} v_{b2}}$ is a randomly fluctuating function of the reactant state energy. Nevertheless, the convolution of the bending and rotational motions has the effect of smoothing out these apparent fluctuations: of course, the reaction probability $k_{v_{b1} v_{b2} v_{\text{bend}} J M_J}$ still varies erratically as a function of the reactant state

²We have discussed previously (8) the conceptual difficulty associated with the state-density near the zero point energy in eq. [10], which does not arise in eq. [7].

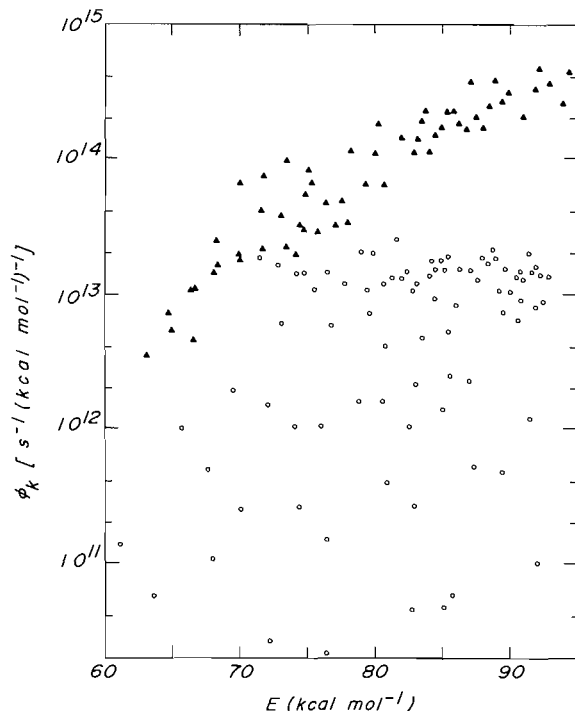


FIG. 1. Grain reaction probability function ϕ_k for the dissociation of N_2O calculated using eq. [11b] with $\Delta E = 0.05$ kcal mol $^{-1}$. E is the energy of the reactant state relative to the zero point energy. Open circles: bending and rotational motions excluded, eq. [14a]; solid triangles: bending motion included, eq. [14b].

energy, but now the enormous increase in the reactant state density means that over a typical energy range of ΔE (~ 0.1 kcal mol $^{-1}$), the *apparent* reaction probability distribution is smoothed out considerably. This is illustrated in Fig. 1 for the N_2O reaction, where we sum up the reaction probabilities from states in energy grids of 0.05 kcal mol $^{-1}$. When neither the bending nor the rotational motions are included, i.e.

$$[14a] \quad \phi_k = \{\sum g_{v_{b1} v_{b2} 0 0} k_{v_{b1} v_{b2} 0 0}\} / \Delta E$$

where the summation is over the $(v_{b1}, v_{b2}, 0, 0)$ states within E_k and E_{k+1} , the grain reaction probability function ϕ_k exhibits random fluctuations (open circles). On the other hand, as soon as the bending motions are included, i.e.

$$[14b] \quad \phi_k = \{\sum g_{v_{b1} v_{b2} v_{bend} 0} k_{v_{b1} v_{b2} v_{bend} 0}\} / \Delta E$$

the summation being over the $(v_{b1}, v_{b2}, v_{bend}, 0)$ states within E_k and E_{k+1} , the smoothing of ϕ_k immediately becomes apparent (filled triangles: note that in this case, only a representative fraction of data points is shown in the figure for the sake of

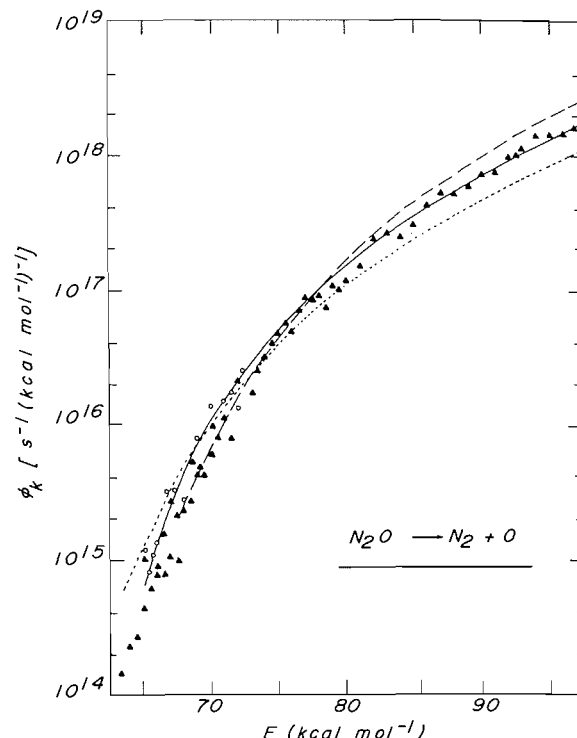


FIG. 2. Grain reaction probability function ϕ_k for the dissociation of N_2O , with $\Delta E = 0.05$ kcal mol $^{-1}$. E is the energy of the reactant state relative to the zero point energy. Solid triangles: exact probabilities from eq. [14c]; solid curve: approximation from inversion of Arrhenius rate law with $A_\infty = 1.41 \times 10^{12}$ s $^{-1}$ and $E_\infty = 65.10$ kcal mol $^{-1}$, eq. [15]; dotted curve: eq. [15a] with $A_\infty' = 7.05 \times 10^{11}$ s $^{-1}$ and $E_\infty' = 63.25$ kcal mol $^{-1}$; dashed curve: eq. [15b] with $A_\infty'' = 2.82 \times 10^{12}$ s $^{-1}$ and $E_\infty'' = 67.87$ kcal mol $^{-1}$. The open circles indicate fluctuations of $\phi_{k,app}$ near the threshold.

clarity). The further smoothing of ϕ_k with the convolution of the rotational states, i.e.

$$[14c] \quad \phi_k = \{\sum g_{v_{b1} v_{b2} v_{bend} J M_J} k_{v_{b1} v_{b2} v_{bend} J M_J}\} / \Delta E$$

the summation being over the $(v_{b1}, v_{b2}, v_{bend}, J, M_J)$ states within E_k and E_{k+1} , is shown by the quasi-smooth array of solid triangles in Fig. 2. Similar results are obtained for the CO_2 reaction; they will not be presented here.

IV. The "Arrhenius" Reaction Probability Function

Given the relevant spectroscopic parameters (Table 1 of ref. 1), it is a simple matter to calculate the reactant state-sum $G(E)$ and, combining eqs. [7] and [11b], the grain reaction probability function $\phi_{k,app}$,

$$[15] \quad \phi_{k,app} = A_\infty [G(E_{k+1} - E_\infty) - G(E_k - E_\infty)] / \Delta E$$

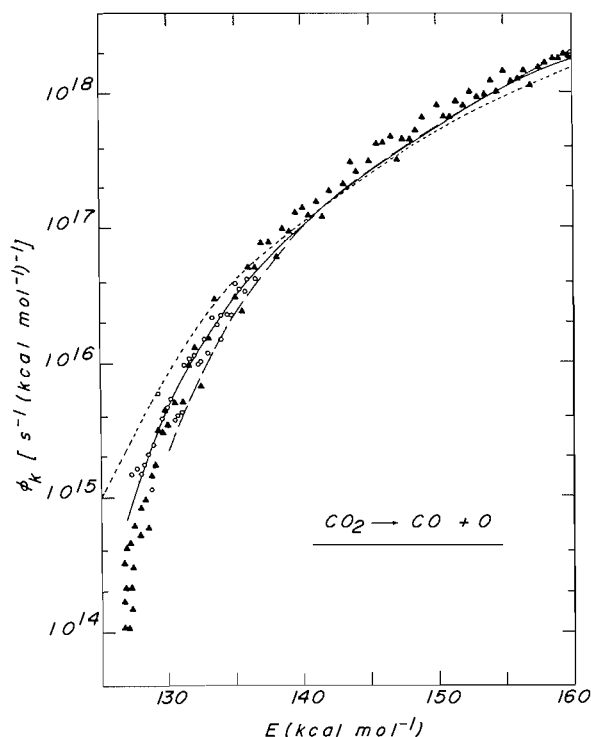


FIG. 3. Grained reaction probability function ϕ_k for the dissociation of CO_2 with $\Delta E = 0.05 \text{ kcal mol}^{-1}$. See captions in Fig. 2: $A_\infty = 1.80 \times 10^{12} \text{ s}^{-1}$, $E_\infty = 127.3 \text{ kcal mol}^{-1}$; $A_{\infty'} = 1.20 \times 10^{12} \text{ s}^{-1}$, $E_{\infty'} = 124.8 \text{ kcal mol}^{-1}$; $A_{\infty''} = 2.70 \times 10^{12} \text{ s}^{-1}$ and $E_{\infty''} = 129.8 \text{ kcal mol}^{-1}$.

Equation [15] has been calculated using the Arrhenius parameters obtained in ref. 1 and compared with ϕ_k in Fig. 2 for N_2O and in Fig. 3 for CO_2 . In both figures, the values of ϕ_k are represented by solid triangles, while $\phi_{k,\text{app}}$ is shown as a smooth curve: near the threshold, the $\phi_{k,\text{app}}$ values actually exhibit some fluctuations, as indicated by the open circles. The remarkable agreement between the solid curve and the triangles in the intermediate energy range (between 10 and 30 kcal mol^{-1} above the Arrhenius activation energy) should not be surprising: the strict Arrhenius law approximates the exact rates best at the medium temperatures (1000–3000 K in N_2O and 2000–5000 K in CO_2) where the reactive state population distribution (i.e., $F(E_j) \exp(-\beta E_j)$ in eq. [4]) peaks within this energy range. It is interesting to note that the high energy region of ϕ_k is important only at very high temperature, where the concept of an infinite pressure rate constant (in the Tolman sense (9)) becomes less well-defined (8). Conversely, the low energy region (near the threshold) of ϕ_k is important only at very low temperature (for recombination studies); hence we may expect the largest departure between ϕ_k and $\phi_{k,\text{app}}$ in this energy region. A flow chart illustrating the principal computational steps in this comparison is given in Fig. 4.

Figures 2 and 3 show that, at least for the two reaction systems studied here, the inversion of the strict Arrhenius rate law (eqs. [3] and [7]) does lead to a reasonable approximation to the exact reaction

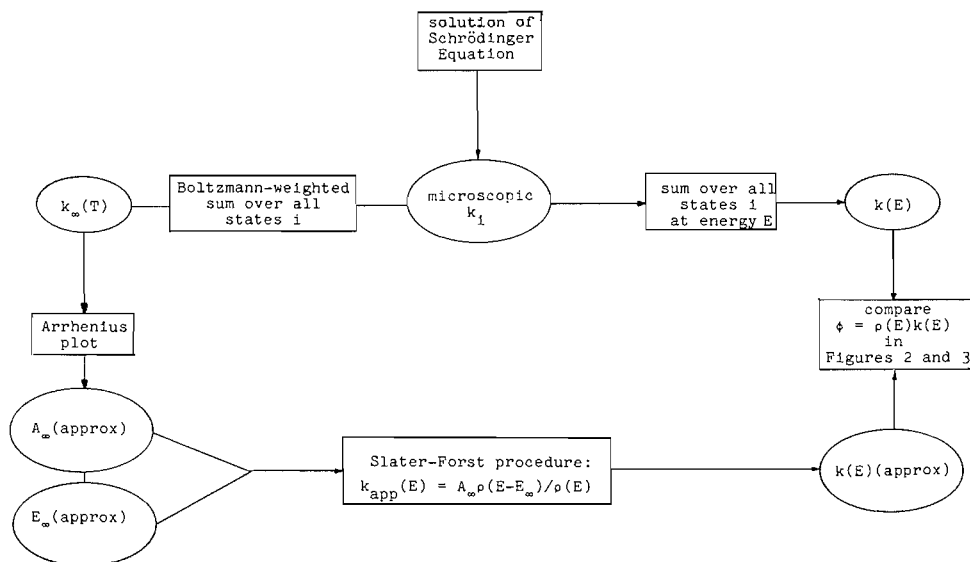


FIG. 4. Schematic flow chart of the test procedure.

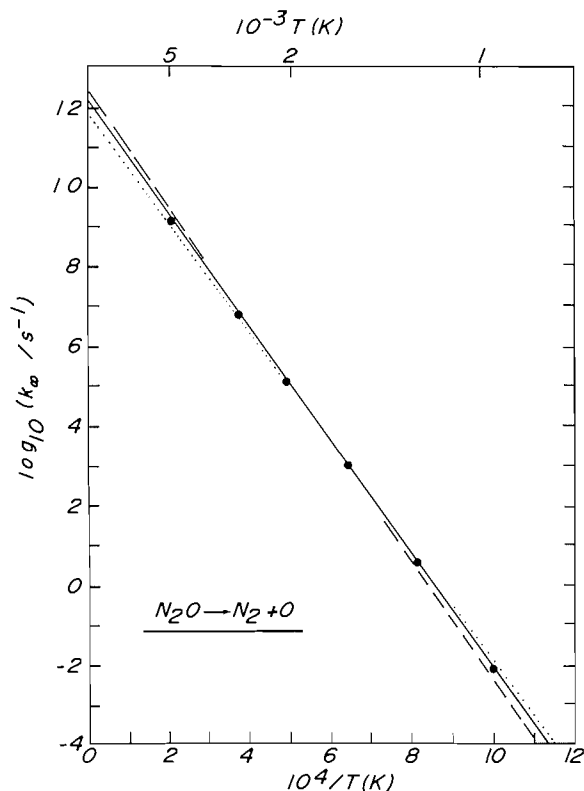


FIG. 5. Arrhenius plot of the infinite pressure rate constant for the dissociation of N_2O . Solid circles: computed (exact) rates from ref. 1; solid line: Arrhenius rate law with $A_\infty = 1.41 \times 10^{12} \text{ s}^{-1}$ and $E_\infty = 65.10 \text{ kcal mol}^{-1}$; dotted line: Arrhenius rate law with $A_\infty' = 7.05 \times 10^{11} \text{ s}^{-1}$ and $E_\infty' = 63.25 \text{ kcal mol}^{-1}$; dashed line: Arrhenius rate law with $A_\infty'' = 2.82 \times 10^{12} \text{ s}^{-1}$ and $E_\infty'' = 67.87 \text{ kcal mol}^{-1}$.

probability function. It is important to assess the sensitivity of the inversion procedure to variations of the Arrhenius parameters. For each reaction, two additional pairs of parameters have been chosen to test this sensitivity. For N_2O , $A_\infty' = 7.05 \times 10^{11} \text{ s}^{-1}$, $E_\infty' = 63.25 \text{ kcal mol}^{-1}$ and $A_\infty'' = 2.82 \times 10^{12} \text{ s}^{-1}$, $E_\infty'' = 67.87 \text{ kcal mol}^{-1}$; for CO_2 , $A_\infty' = 1.20 \times 10^{12} \text{ s}^{-1}$, $E_\infty' = 124.8 \text{ kcal mol}^{-1}$ and $A_\infty'' = 2.70 \times 10^{12} \text{ s}^{-1}$, $E_\infty'' = 129.8 \text{ kcal mol}^{-1}$. They are compared with the "best" parameters ($A_\infty = 1.41 \times 10^{12} \text{ s}^{-1}$ and $E_\infty = 65.1 \text{ kcal mol}^{-1}$ for N_2O and $A_\infty = 1.80 \times 10^{12} \text{ s}^{-1}$ and $E_\infty = 127.3 \text{ kcal mol}^{-1}$ for CO_2) and the exact reaction rates in Figs. 5 (for N_2O) and 6 (for CO_2).

Going back to Figs. 2 and 3, the dotted curves denote the grained reaction probability functions $\phi_{k,\text{app}}'$ obtained using A_∞' and E_∞' :

$$[15a] \quad \phi_{k,\text{app}}' = A_\infty' [G(E_{k+1} - E_\infty') - G(E_k - E_\infty')]/\Delta E$$

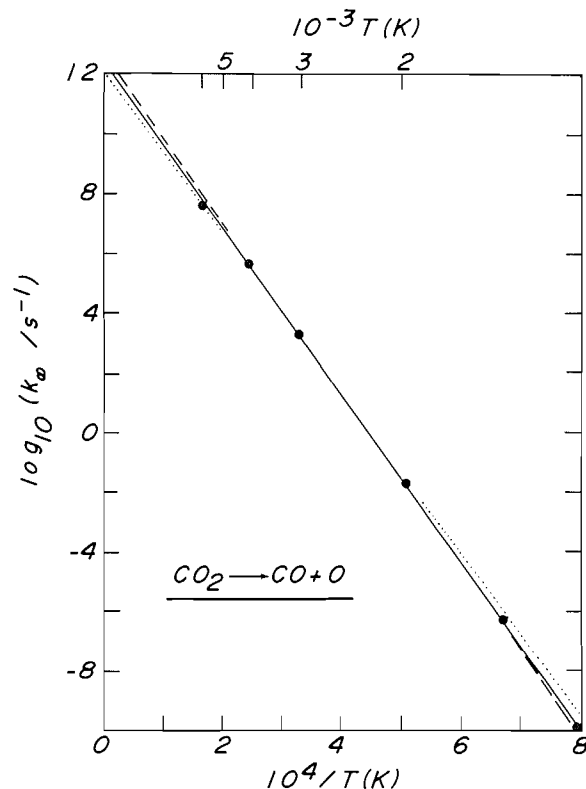


FIG. 6. Arrhenius plot of the infinite pressure rate constant for the dissociation of CO_2 . Solid circles: computed (exact) rates from ref. 1; solid line: Arrhenius rate law with $A_\infty = 1.80 \times 10^{12} \text{ s}^{-1}$ and $E_\infty = 127.3 \text{ kcal mol}^{-1}$; dotted line: Arrhenius rate law with $A_\infty' = 1.20 \times 10^{12} \text{ s}^{-1}$ and $E_\infty' = 124.8 \text{ kcal mol}^{-1}$; dashed line: Arrhenius rate law with $A_\infty'' = 2.70 \times 10^{12} \text{ s}^{-1}$ and $E_\infty'' = 129.8 \text{ kcal mol}^{-1}$.

the dashed curves denote $\phi_{k,\text{app}}''$ obtained from A_∞'' and E_∞'' :

$$[15b] \quad \phi_{k,\text{app}}'' = A_\infty'' [G(E_{k+1} - E_\infty'') - G(E_k - E_\infty'')]/\Delta E$$

In both figures, these two curves intersect with the solid curve ($\phi_{k,\text{app}}$, eq. [15]) and with each other in the intermediate energy region. In other words, both $\phi_{k,\text{app}}'$ and $\phi_{k,\text{app}}''$ approximate the exact function best in this energy region. This is to be expected since both sets of Arrhenius parameters approximate the exact rates k_∞ best in the medium temperature range. As we have discussed previously (8), this is simply a consequence of the linearity property of the Laplace Transform.

Likewise, the divergent departure between $\phi_{k,\text{app}}'$ and $\phi_{k,\text{app}}''$ at very high energy is related to the corresponding departure between the respective Arrhenius expressions at very high temperature.

Here, it is pleasing to note that in the case of N_2O (Fig. 2), a factor of 4 variation in the pre-exponential factor results in only a factor of 2 variation of $\phi_{k,app}$ at 30 kcal mol⁻¹ above E_∞ . In experiments where a factor of 2 uncertainty in the A -factor is not atypical, the variation of $\phi_{k,app}$ in the corresponding energy region may be expected to be even smaller, as is illustrated in the case of CO_2 (Fig. 3).

Discrepancies between the respective $\phi_{k,app}$'s are most drastic near E_∞ , but even here, the difference in absolute magnitude between $\phi_{k,app}'$ and $\phi_{k,app}''$ at and above E_∞'' is not phenomenal (a factor of 2 at E_∞'' for a variation of 4.6 kcal mol⁻¹ in the Arrhenius activation energy in N_2O ; a factor of 3 at E_∞'' for a variation of 5 kcal mol⁻¹ in the Arrhenius activation energy in CO_2). Notice that even in this energy region, $\phi_{k,app}$ gives the best approximation to ϕ_k . The most serious difficulty is that $\phi_{k,app}$ is zero between the critical energy of reaction E_0 and the activation energy E_∞ ($E_\infty > E_0$) whereas ϕ_k is not. Note, however, that $\phi_{k,app}$ is *not* zero at E_∞ ; also, ϕ_k is small between E_0 and E_∞ . Hence, the discrepancy between ϕ_k and $\phi_{k,app}$ in this region is unlikely to magnify in rate calculations of typical physical systems (unless the latter sample predominantly this energy region). From a computational standpoint, we may categorise reaction systems into two classes: those in which ($E_\infty - E_0$) is large (such as N_2O), where ϕ_k is extremely small between E_∞ and E_0 (see ref. 1) and may therefore be well approximated by zero, and those in which E_0 and E_∞ are close to one another (such as CO_2), where simple procedures may be employed to interpolate between the zero value of $\phi_{k,app}$ at E_0 and the non-zero values of $\phi_{k,app}$ at and above E_∞ . Alternatively, the overall fit to ϕ_k may be improved by a linear combination of more than one Arrhenius expression as suggested previously (8).

Conclusions

The present work represents a first attempt to test the sensitivity and reliability of the Slater-Forst procedure for inverting the strict Arrhenius rate law using purely theoretical (and realistic) rate data (1); at a later date, similar tests should be made for some adiabatic reactions, but we see no reason to expect that the results will be any less encouraging since the non-adiabatic reactions can be expected, in general, to be more non-Arrhenius. For the two reactions studied, the test results are indeed positive in the sense that (i) given the "best" Arrhenius parameters, the inversion procedure yields faithfully

a very good approximation to the exact probability function. It is important to recognise the nature of the test we are making: if we simply take the exact computed $k_\infty(T)$ function and perform an inverse Laplace Transform, then we should (barring rounding errors) recover the original $k(E)$; however, the function upon which we apply the inverse Laplace Transform is not $k_\infty(T)$, but an approximate parametrisation of it, viz. $k_\infty(T) = A_\infty \exp(-E_\infty/RT)$. Moreover, by assuming that E_∞ can only be determined to within ± 2 or 3 kcal mol⁻¹, we have shown that (ii) reasonably approximate (experimental) rate laws would lead to reasonable approximations to the exact reaction probability function $k(E)$. It should also be noted that we are guaranteed approximate coincidence between the original theoretical and the doubly transformed values of $k(E)$ around the mean reaction energy, but what is remarkable about Figs. 2 and 3 is the fact that (iii) for these two reactions the inverse Laplace Transform approximation is uniformly satisfactory at all energies above the activation energy E_∞ , including the very high energies which make little contribution to the reaction rate in the temperature ranges considered (1). Finally, it would appear that if it proves necessary (iv) simple interpolation procedures should prove adequate for synthesising approximate values of $k(E)$ between E_∞ and E_0 (the critical energy of the reaction), both when E_∞ and E_0 are far apart (e.g. N_2O) and when they are close to one another (e.g. CO_2).

Acknowledgements

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